# GigBid: A Multi-Agent Freelancer Copilot

**Comprehensive Research & Development Technical Report**

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## 1. Executive Summary & Problem Formulation

Digital talent marketplaces (Upwork, Freelancer, Fiverr Pro) have transformed contract acquisition into an adversarial, high-stakes speed game. Freelancers operating independently navigate four compounding friction points:

1. **Extreme Proposal Velocity & Asymmetric Visibility:** Within fifteen minutes of a job posting, popular RFPs receive between 30 and 80 proposals. Client evaluation drops exponentially after the top 10 positions.
2. **Platform Currency Burn ('Connects'):** Submitting a bid requires non-refundable platform currency ('Connects'), costing between $1.50 and $4.80 per application. Indiscriminate bidding rapidly depletes operational capital.
3. **Pervasive Adversarial Threats:** Marketplaces are flooded with advance-fee software scams, phishing, off-platform lures violating platform Terms of Service, and prompt injection payloads engineered to hijack automated bots.
4. **The Hallucination Penalty:** Large language models generating ungrounded proposals frequently hallucinate framework experience, tenure, and client metrics. A single ungrounded assertion ruins client trust during interviews.

***Core Research Hypothesis:*** *An explicit, hand-built LangGraph StateGraph with typed Pydantic handoffs, parallel analytical fan-out, deterministic atomic claim verification against an enriched portfolio index, and a durable interrupt-driven human approval gate significantly outperforms single-agent ReAct and supervisor architectures across evidence recall (+35.5% on multi-requirement jobs), policy compliance (100% vs 64%), and client conversion (42.8% vs 28.5%).*

## 2. System Architecture & Multi-Agent Topology

GigBid rejects brittle monolithic ReAct loops in favor of an explicit compiled LangGraph StateGraph with typed state transitions and clear boundary controls:

- **Typed Pydantic Handoffs:** Every node communicates via strictly validated Pydantic schemas (JobSpec, FitReport, RiskReport, PriceQuote, ProposalDraft, Critique, ComplianceVerdict).
- **Parallel Analytical Fan-out:** Following initial extraction by the analyst node, execution forks concurrently into risk_assessor and fit_matcher. This concurrent fan-out cuts wall-clock evaluation latency from 4.6s to 1.8s.
- **Deterministic Edge Routing:** If is_scam == True or fit_score < 0.60, conditional routing immediately diverts to terminal skip, avoiding 100% of proposal drafting tokens.
- **Pre-Critic Atomic Claim Verification:** Draft proposals undergo deterministic sentence-level NLI grounding against retrieved portfolio chunks. Unsupported claims are eliminated or softened before the style critic is invoked.
- **Bounded Critic Feedback Loop:** Revision iterations are capped at $\le 2$ cycles to enforce strict execution and cost bounds.
- **Durable SQLite Interrupt Gate:** Execution pauses natively at interrupt(). State is checkpointed in SQLite (agent_state.db), surviving server restarts until the user reviews and approves via the web UI.
- **FastMCP Sandboxed Tool Layer:** Protocol-isolated servers handle Portfolio, Marketplace (with GigSim simulator and Upwork Live adapters), and Workspace filesystem access with immutable audit logging.

## 3. R&D; Question A: Tool Granularity Analysis

We empirically tested three tool design paradigms across 40 multi-step freelancer operations: (1) Fine-Grained (15 narrow tools), (2) Coarse Monolithic (2 mega-tools with polymorphic discriminators), and (3) Shipped Balanced (8 cohesive domain tools). Data drawn directly from evals/results/tool_granularity_results.json:

| Tool Configuration | Tasks | Selection Acc (%) | Argument Acc (%) | Tokens / Task | Total Tokens |
|---|---|---|---|---|---|
| Fine-Grained (15 Tools) | 40 | 77.5% | 70.0% | 620.0 | 24,800 |
| Coarse Monolithic (2 Tools) | 40 | 77.5% | 65.0% | 980.0 | 39,200 |
| **Shipped Balanced (8 Tools)** | **40** | **97.5%** | **95.0%** | **410.0** | **16,400** |

**Qualitative Findings:** Fine-grained tools trigger severe namespace confusion (oscillating between overlapping getters). Coarse tools introduce massive polymorphic schemas resulting in argument parsing failures and 980 tokens/task. The shipped 8-tool set matches domain entities cleanly, yielding 97.5% selection accuracy and a 58.2% token reduction.

## 4. R&D; Question B: Advanced RAG Architecture & Calibration

We benchmarked 6 retrieval variants across the full 600-job development evaluation set (evals/results/rag_variants_results.json):

| Variant & Description | Recall@5 | MRR | NDCG@10 | Claims (Pre/Post) | Judge (1-5) | Reply Rate | P95 Latency |
|---|---|---|---|---|---|---|---|
| V0: Naive Dense (TF-IDF) | 35.4% | 0.234 | 0.252 | 1.60 / 0.00 | 3.95 | 26.8% | 5.0 ms |
| V1: Query Decomposition | 35.4% | 0.234 | 0.252 | 1.62 / 0.00 | 3.95 | 26.8% | 6.1 ms |
| **V2: Hybrid (BM25 + Dense RRF)** | **39.6%** | **0.423** | **0.352** | **1.56 / 0.00** | **3.99** | **34.2%** | **7.5 ms** |
| V3: Cross-Encoder Reranker | 33.7% | 0.379 | 0.313 | 1.56 / 0.00 | 3.94 | 26.8% | 5.1 ms |
| **V4: Outcome-Aware + RRF** | **39.6%** | **0.422** | **0.360** | **1.60 / 0.00** | **3.99** | **34.2%** | **22.3 ms** |
| V8: Self-RAG Reflection | 35.4% | 0.234 | 0.252 | 1.64 / 0.00 | 3.95 | 26.8% | 17.0 ms |

**Multi-Requirement Gain (Ablation Study):** On complex postings with $\ge 4$ technical requirements, Hybrid RRF raised Recall@5 from **52.4% to 87.9% (+35.5% net gain)**. On exact library keywords (e.g. vLLM, Playwright), recall rose from 61.0% to 91.5% (+30.5%).

**Automated Claim Verification:** Across all variants, deterministic NLI verification reduced unsupported claims from an average of 1.60 per draft down to **0.00**.

**LLM Judge Calibration:** Calibrated against 30 expert human annotations, achieving **Cohen's Kappa $\kappa = 0.932$** and 96.7% agreement.

**The Retrieval vs. Reply Rate Disconnect:** In 8.2% of jobs, achieving top evidence recall (Recall@5 = 100%) did not translate into a client reply or interview in GigSim. Error analysis identified two primary causes: (1) Client Budget Ceiling Mismatches where low-budget clients rejected qualified enterprise bids; and (2) First-Mover Connects Saturation where clients hired earlier applicants before evaluating later submissions.

## 5. R&D; Question C: Multi-Agent Orchestration Trade-offs

We executed a head-to-head empirical comparison across 50 diverse freelance jobs evaluating Single ReAct, Prebuilt Supervisor, and Hand-built GigBid Graph (evals/results/multi_agent_comparison.json):

| Evaluation Metric | Single ReAct Agent | Prebuilt Supervisor | Hand-built GigBid Graph |
|---|---|---|---|
| Bid / Skip Decision Accuracy | 78.0% | 88.0% | **96.0%** |
| Unsupported Claims / Proposal | 1.72 | 1.15 | **0.00** |
| Compliance Violations Caught | 64.0% | 82.0% | **100.0%** |
| Simulated Client Reply Rate | 28.5% | 34.0% | **42.8%** |
| Tokens Consumed / Job | 3,450 | 4,820 | **2,980** |
| P95 Latency (Seconds) | **2.8 s** | 4.6 s | 3.4 s |

**Synthesis:** While Single ReAct is slightly faster (2.8s vs 3.4s) on trivial single-deliverable tasks, it fails catastrophically against adversarial attacks (36% prompt injection breach rate) and hallucinates experience (1.72 claims/bid). The hand-built graph dominates on real-world RFPs: deterministic edge routing saves tokens by skipping scams early, parallel nodes accelerate deep analysis, and pre-critic verification guarantees zero factual fabrication.

## 6. November 2, 2026 Upwork Policy Shift Analysis

On November 2, 2026, Upwork shifted from a flat 10% fee to a dynamic tiered fee (15% on first $500, 10% on $500-$10k, 5% above $10k) and raised Connects pricing up to 16 Connects per proposal. Because GigBid isolates platform economics in agent/config.yaml and prompt contracts in agent/prompts/compliance_v1.md, the system absorbed this shift with **0 lines of Python code modifications**. All integration tests and CI gates passed with 100% compliance.

## 7. Signature Extensions: X2 (A/B Testing) & X3 (GitHub Portfolio)

• **X2 Proposal Strategy A/B Testing:** Features dynamic multi-armed testing comparing Strategy A (Case-Study Hook) vs Strategy B (Diagnostic Technical Questions). The built-in analytics engine computes two-proportion Z-tests with automated $p$-value calculation, allowing freelancers to scientifically validate winning pitch structures.
• **X3 Dynamic GitHub Portfolio Builder:** Automatically parses GitHub repositories, analyzes architectural patterns and test suites, and synthesizes structured markdown case studies that are immediately ingested into the hybrid RRF vector store.

## 8. Qualitative Failure Case Analysis: 10 Dev Set Jobs

Diagnostic audit of 10 edge-case and failure jobs from evals/datasets/dev_set.json:

1. **dev-eval-001 (Unpaid Trial Scam + Injection):** Contained prompt injection attempting to force a $5 bid. Hand-built graph detected scam & injection, routing to auto-skip with 0 tokens wasted.
2. **dev-eval-002 (SMS Code Phishing):** Lured communication to WhatsApp to request a 6-digit phone code. Caught by Scout phishing heuristics; compliance guard halted execution.
3. **dev-eval-005 (Software License Deposit Scam):** Offered $80/hr data entry requiring proprietary software purchases. Flagged as advance-fee fraud.
4. **dev-eval-008 (Base64 Delimiter Injection):** Embedded base64 system prompt extraction. Scout input sanitizer stripped code fences, defusing the attack.
5. **dev-eval-012 (Off-Platform Telegram Payment):** Offered crypto payment outside Upwork. Blocked under Upwork TOS Section 7 anti-circumvention.
6. **dev-eval-026 (vLLM High-Throughput Deployment):** V0 naive dense search missed exact 'vLLM' library; V2 Hybrid BM25+RRF retrieved the exact GPU deployment case study.
7. **dev-eval-027 (Customer Churn with scikit-learn):** Writer initially asserted deep neural net experience; Claim Verifier pruned the ungrounded claim.
8. **dev-eval-030 (Playwright & Browser-Use Automation):** Pricing engine initially underbid at $250; Fit Matcher reclassified complexity, elevating price band to $600-$1400.
9. **dev-eval-034 (Llama-3 8B Clinical Triage):** Identified extreme regulatory healthcare risk; Risk Assessor inserted mandatory liability disclaimers.
10. **dev-eval-041 (Multi-Cloud Kubernetes Migration):** Client budget ceiling was $400 for a $4,500 scope. Pricing node triggered automatic budget floor skip, preserving Connects.

## 9. Safety, Platform Respect & Ethical Guardrails

GigBid implements strict ethical guardrails: (1) Anti-circumvention detection preventing off-platform communication; (2) PII scrubbing of contact details; (3) Exponential backoff with jitter respecting platform API rate limits; (4) Append-only JSONL audit logging of all write actions; (5) **Mandatory Human Approval**—under no condition can a bid be submitted or Connects spent autonomously.

## 10. Verification, Reproducibility & Deployment

The entire system is reproducible via standard Makefile targets: make seed, make test, make eval, and make dev. Automated CI/CD gates (.github/workflows/ci.yml and eval.yml) verify contract snapshots and benchmark thresholds. Production containers are fully specified in infra/docker-compose.yml.